🏛️ Credit Risk & Recovery Strategy Lab

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Target Stakeholders: Recovery Operations, Risk Modelers, 2nd Line Risk Strategy, & Chief Credit Officers (CCO)

📊 Executive Summary

The **Credit Risk & Recovery Strategy Lab** is a repository of ideas for low-code/no code financial engineering and predictive modeling tools designed to optimize items such as the **Net Present Value (NPV)** of delinquent portfolios. The majority of the items in the repository are designed to bridge the gap between 1st-line operational capacity and 2nd-line risk appetite. The idea behind this suite and the lab is to provide quantitative frameworks that can be used to solve various issues related to the disposition and recovery of Non-Performing Loan (NPL) portfolios (e.g the "Hold vs. Liquidate" dilemma).

Strategic Business Value

- **Capital Efficiency (CET1):** Several models use PD/LGD/EAD matrices to model inherent loss, directly informing capital buffer requirements and RWA reduction.
- **Yield Optimization:** The folder contains DCF and NPV models to identify the "Crossover Point" where internal recovery efforts outperform external debt sales.
- **Behavioral Segmentation:** Models that eplace broad collections with "Propensity to Pay" (PTP) scoring, ensuring high-touch resources are allocated to high-yield targets.

🚀 Strategic Modules & Decision Engines

The repository is structured into functional "Labs" that mirror the lifecycle of a delinquent account.

🧠 Propensity & Behavioral Segmentation

- **Goal:** Differentiate "Will-Pay" from "Won't-Pay" borrowers using advanced classification.

- **Core Assets:** `PTP_Model.ipynb`, `PropPay_LogReg_Model.ipynb`, `Comm_Pref_Segmentation.ipynb`, `KNN_CreditModel.ipynb`
- **Impact:** Enables precision communication (e.g., SMS vs. Legal) based on historical payment velocity and contact preference.

⚖️ Financial Engineering & Liquidation Strategy

- **Goal:** Determine the most profitable recovery path through time-value-of-money (TVM) analysis.
- **Core Assets:** `NPV_Liquidation.ipynb`, `DCF_Model.ipynb`, `Liquidation_Decision_Tool.ipynb`, `Cure_v_Roll_Rate.ipynb`
- **Impact:** Models the "Yield Premium" of holding accounts vs. the immediate cash-flow benefit of bulk asset sales.

📉 Risk Exposure & Loss Modeling

- **Goal:** Quantify the "Inherent Loss" within credit card and retail portfolios.
- **Core Assets:** `EAD_PD_LGD_Matrix.ipynb`, `PD_LGD_Model.ipynb`, `InherentLossCreditCard.ipynb`, `CreditClassification.ipynb`
- **Impact:** Provides 2nd-line defense with a structured view of Probability of Default (PD) and Loss Given Default (LGD) quadrants.

📍 Visual Analytics & Risk Matrices

- **Goal:** Provide "At-a-Glance" portfolio health snapshots for risk committees.
- **Core Assets:** `BubbleRiskDollarsDisplayed.ipynb`, `CCardCollectionMatrixMonths.ipynb`, `BubbleMonthsDelinquent.ipynb`
- **Impact:** Visualizes risk concentration across delinquency buckets to identify emerging "bubbles" before they impact the bottom line.

🛠️ Technical Implementation

This lab demonstrates proficiency in full-stack financial data science and automated modeling.

Layer	Technical Stack
Data Science	Pandas, NumPy, Yfinance (Market data integration)
Machine Learning	Scikit-Learn (Logistic Regression, KNN, Classification)
Financial Logic	Custom DCF/NPV scripts for recovery yield premiums
Visualization	Matplotlib, Seaborn (Bubble charts and risk heatmaps)

📊 Strategic KPI Glossary

The models within this lab are aligned with 2026 Regulatory Compliance and banking standards:

- **Cure vs. Roll Rate:** The velocity at which delinquent accounts return to "Current" status versus rolling into deeper delinquency.
- **EAD (Exposure at Default):** The predicted dollar amount at risk at the moment of default.
- **Operational Capacity Yield:** A unique metric balancing team bandwidth against expected recovery dollars.
- **Inherent Loss:** The modeled loss expected within a segment before any recovery actions are taken.

💻 Installation & Usage

To run these models locally and simulate your own portfolio behaviors:

1. **Clone the repository:**

```
git clone https://github.com
```

Screenshots

Specific output screen shots are contained in each file. Each file provided has the Python code that can be run in Jupyter Notebook, VS Studio, Zed or any Integrated Development Environment (IDE) code editor. At the bottom of each file is the output of the code that is written above it.